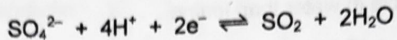
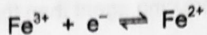
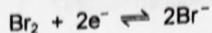
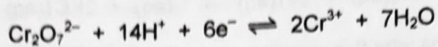


- 16 The dichromate(VI) ion, $\text{Cr}_2\text{O}_7^{2-}$, is a strong oxidising agent in acidic solution. It oxidises bromide ions, iron(II) ions and sulfur dioxide. The relevant half-equations are given.



Which statements about the reactions of 1 mole of $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ are correct?

- 1 It reacts with an excess of bromide ions producing 6 moles of bromine molecules.
- 2 It reacts with 6 moles of iron(II) ions, producing 6 moles of iron(III) ions.
- 3 It reacts with 3 moles of sulfur dioxide, producing 3 moles of sulfate ions.

A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only